

Probability Theory & Statistics

Kahlan AL-Juburi

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About Me

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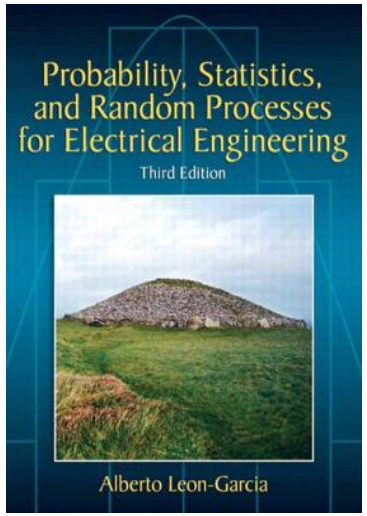
Research: Performance Analysis

Queueing Theory

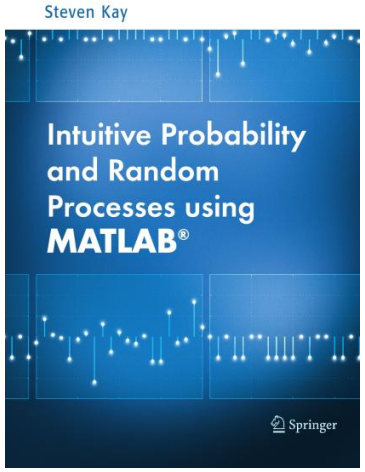
Computer and Communications Networks

Web: Kahlan Aljobory - Google Scholar

Resources



Probability, Statistics, and Random Processes for Electrical Engineering
Albert Leon-Garcia
Pearson, 3rd Edition



Initiative Probability and Random Processes Using MATLAB
Steven Kay
Springer

Course Goals

Students will:

- Learn Foundation of Probability theory.
- Understand how to use random variables.
- Learn elementary probabilities.
- Gain a solid foundation for research (and other courses) involving probabilistic applications.

Course Contents

Module 1: Basic probability theory

(3 weeks)

Module 2: Random variables

(5 weeks)

Module 3: Probability distributions

(5 weeks)

Module 1: Basic Probability Theory

Learning outcomes:

- Student can identify the sample space of a given probabilistic setting.
- Student can differentiate between basic combinatorial techniques and apply them to solve discrete probability problems involving combinatorics.
- Student can derive/prove elementary probabilistic identities and inequalities using Kolmogorov's axioms.
- Student can compute conditional probabilities based on the definition of conditional probability.

Module 1: Basic Probability Theory

Learning outcomes:

- Student can use the Law of Total Probability to express and solve nontrivial probabilities.
- Student can apply Bayes' Rule to compute a priori/a posteriori probabilities.
- Student can determine whether two given events are independent.

Module 2:Random Variables

Learning outcomes:

- Student can define a random variable given a random experiment and/or relevant events.
- Student can differentiate between discrete, continuous and mixed-type random variables.
- Student can compute probabilities of events using cdf/pmf/pdf.
- Student can compute moments of a given random variable given its distribution.

Module 3:Probability Distributions

Learning outcomes:

- Student can compute conditional distributions and moments.
- Student can recognize settings in which certain known distributions (Bernoulli, Binomial, Geometric, Poisson, Uniform, Exponential, Gaussian) can be used.
- Student can infer the parameters of certain known distributions from their distributions, and vice versa.
- Student can derive bounds on probabilities using Markov and Chebyshev inequalities.
- Student can compute joint moments and correlation coefficients given joint distributions.

What You Need to Know

- Elementary logic and set theory
  (and, or, not, intersection, union, set complement, set difference)
- Elementary calculus
  (limits, series and sums, differentiation, integration)
- Matrix algebra (products, determinants, transpose)
- Transforms (optional but helpful)
  (Fourier, Laplace)
- A little programming (optional)

Assessment

Component	Weight (%)
Quizzes Random At the beginning of class, no make-ups	**15**
HomeWorks 3 homeworks no extension AI generated-answers directly rejected	**15**
Midterm Exam 2 exams No make-ups	**10**
Project Random 3-person groups	**10**
Attendance Five absences: fail the class	**0**
Final Exam	**50**

Probability

- What is random?
- What is probability?
- Why do we study probability?
  - Games of chances (Cardano, 1564)
  - Statistical mechanics, behaviour of gases
  - Biological models
  - Economics, actuarial science
  - Logistics and operational research
  - Quantum mechanics
  - Electrical engineering and informatics:

    communications (signal in random environments, noise), computer processor and network performance, signal processing (prediction, estimation), machine learning, data science.

Laplace's rule

- Principle of indifference / Principle of insufficient reason
- Equal priors (a priori probabilities)
- "Probability is the ratio of the *favored events* to the total possible events."

Q: If I roll a fair die, what is the probability that the outcome is

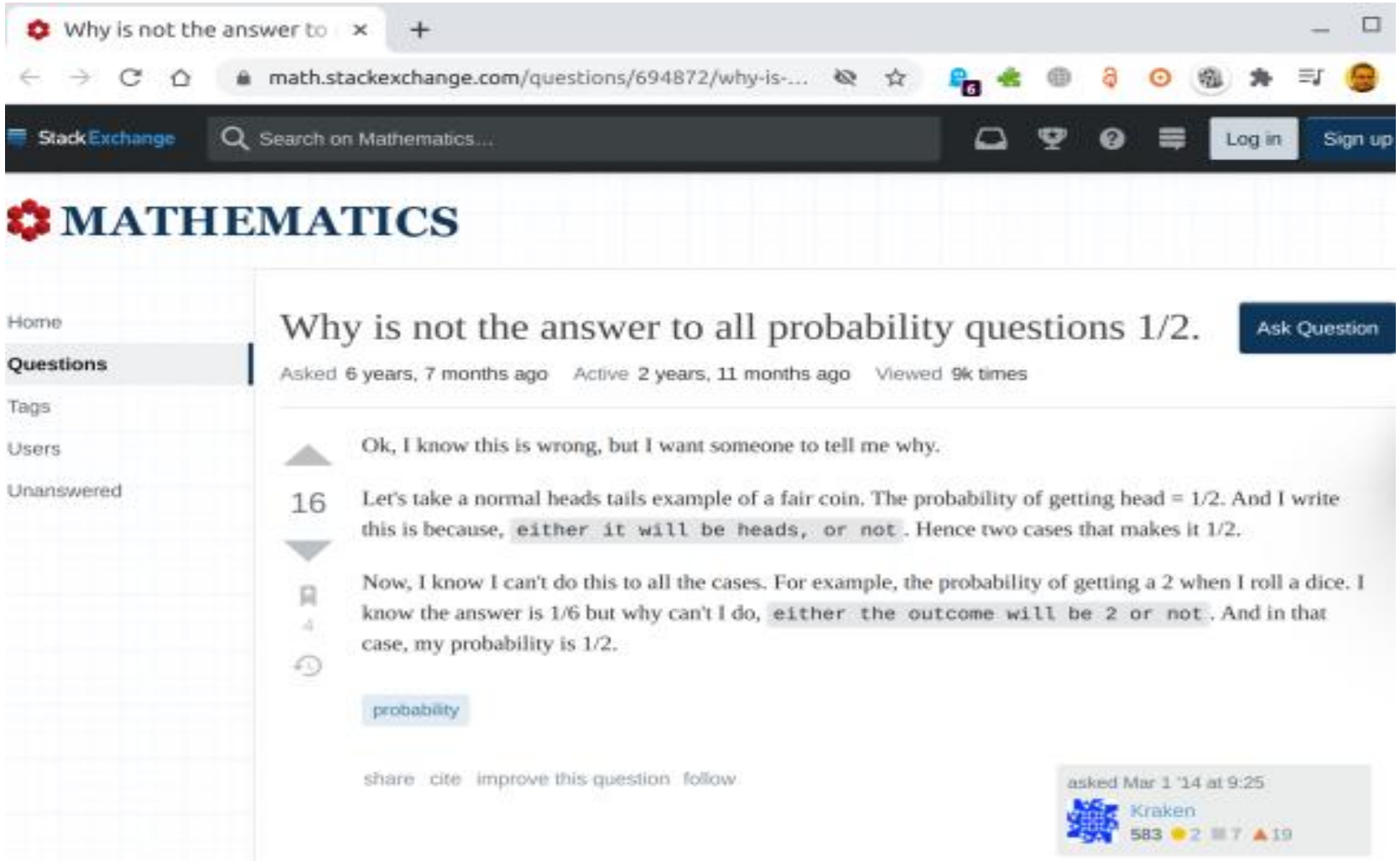
a) 3?

b) Even?

c) Not prime?

d) 7?

Is all probability 1/2 ?



Random Experiments

Sample Space: is the set of all possible outcomes

Q: What is the sample space for the following random experiments?

a) Coin toss
b) 3-coin toss
c) Die throw
d) Temperature measurement
e) Census

Random event: is a subset of the sample space

Counting

Q: Suppose that a student, Muhammed, eats at a café with 10 different dishes for a week, picking his meal completely at random each day.

a) How many possible menus can Muhammed have during these 7 days?

b) What is the probability that Muhammed has the same meal everyday?

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Sampling with replacement with order: There are n^k ways to choose a sequence of ***k*** items out of a population of ***n***.

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c) What is the probability that Muhammed has at most two different kinds of meal?

Counting

d) What if Muhammed wants to eat something different everyday? (how does he pick)

Sampling without replacement with order: There are

$$\prod_{i=1}^{k}(n-i+1) = \frac{n!}{(n-k)!}, \quad n \geq k,$$

ways to choose a sequence of k items out of a population of n, no reuse of items

e) What if Muhammed want to eat something different from what he had the previous day?

Counting

f) Now assume Muhammed, takes three of his friends to his café. Suppose they enter the café one by one, randomly. How many orders are there for their entry?

This is a special case of sampling without replacement with order, and is called a **Permutation: *n!***

Counting

g) Lets say this is a self-service place, and the group decides to appoint one among them to fetch the food. How many different possibilities? Suppose the trays are large, and they decide two people is needed. How many now?

Sampling without replacement without order: choose a ***group*** of ***k*** items out of a population of ***n***, no reuse of items. This is called **combination**.

Counting

Think back to permutations. We can compute the number of permutations of ***k*** out of ***n*** in two ways:

1. $(n)(n-1)\cdots(n-k+1) = \dfrac{n!}{(n-k)!}$
2. Choose a combination k out of n first, then permute them:

$$C_k^n \cdot k! = \frac{n!}{(n-k)!}$$

$$C_k^n = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

"The binomial coefficient"

Counting

Can we generalize the binomial coefficient?

Q: A group of 12 friends wants to play a game with four 3-person teams. How many possible arrangements?

Counting

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What if I need three teams of size 2, 4, and 5, out of 12?

Counting

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What if I need three teams of size 2, 4, and 5, out of 12?

Multinomial coefficients: The number of ways to partition a set of size n into s subsets of sizes $k_1, k_2, \ldots, k_s$, where $\sum_{i=1}^{s} k_i = n$:

$$\frac{n!}{k_1!k_2!\cdots k_s!} = \frac{n!}{\prod_{i=1}^{s} k_i!}$$

Counting

Q: Back to Muhammed and his three friends at the face with 10 dishes, how many different orders are there?

An appropriate question at this point: *From whose perspective?*

a) From Muhammed and friends' perspective?

b) From the cook's perspective?

Sampling with replacement without order: Choose ***k*** items out of population of ***n*** with possible repetition i.e. distribute ***k*** items into ***n*** categories. This is the problem of counting ***multisets***, which is given by the ***multiset coefficients***:

$$\left(\!\binom{n}{k}\!\right) = \binom{k+n-1}{k} = \frac{(k+n-1)!}{(n-1)!k!}$$